

Mathematical Modelling - MAT 1203

Assignment - 10%
Open book Test - 20%
Final Exam - 70%

What is mathematical modelling?

A set of mathematical model that describes the input-output behaviour of a system.

What is mathematical model?

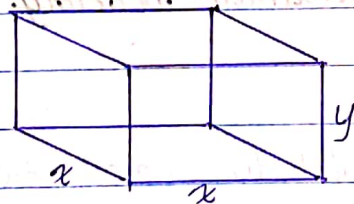
A maths model is when we use mathematical ideas to 'model' an object or situation from the real world.

Mathematical models use estimated measurement to help us be more precise when planning out what we can do with the real version of the model.

We already use mathematical models every day in the classroom. Take the topic of cubic meters and volume.

We can use the equation, width - x , height - y , Length - x , to figure out space inside of a box.
width - x , height - y , length - x

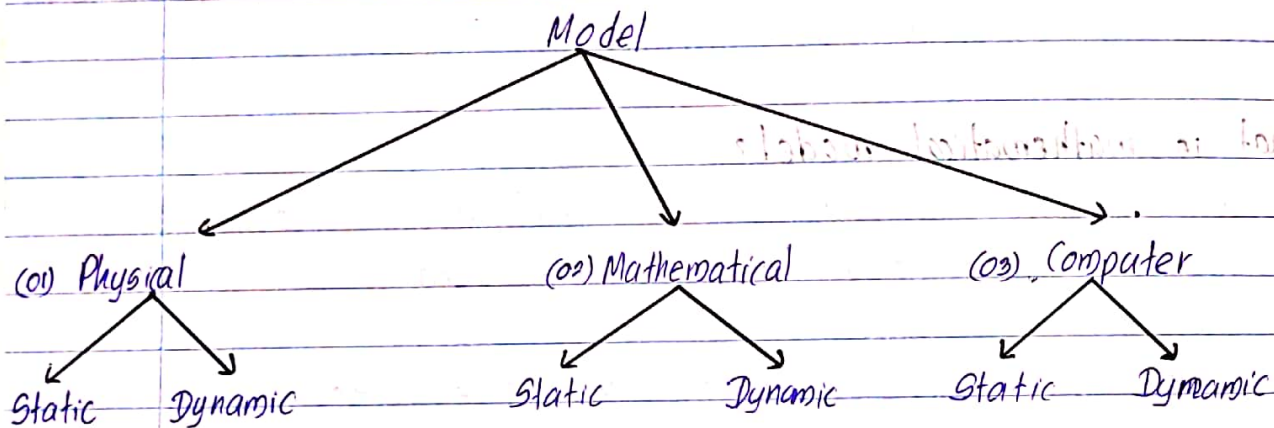
This is useful because it shows us how much we could fit



Model

A model is a simplified representation or concept of reality.

- ★ Reality is generally too complex to copy exactly.
- ★ An examples for Imitation or emulation.
- ★ Much of complexity is actually irrelevant in problem solving.
- ★ A representation used to visualize something.



Types of Systems.

Static System

If a system does not change with time, it is called static system.

Dynamic System

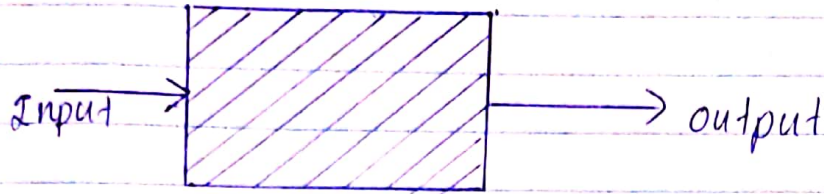
If a system changes with time, it is called dynamic system.

Classification of mathematical models:

- (01) Linear Vs Non-linear.
- (02) Deterministic Vs probabilistic (Stochastic).
- (03) Static Vs Dynamic.
- (04) Discrete Vs Continuous.

Black Box Model

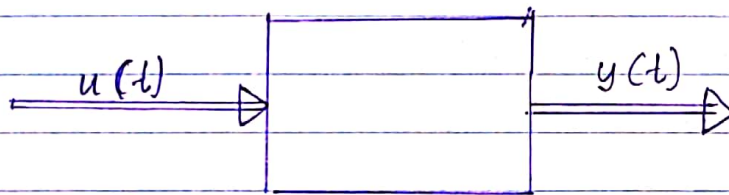
- ★ When only input and output are known.
- ★ Internal dynamics are either too complex or unknown.



- ★ Easy to Model.

Grey Box Model

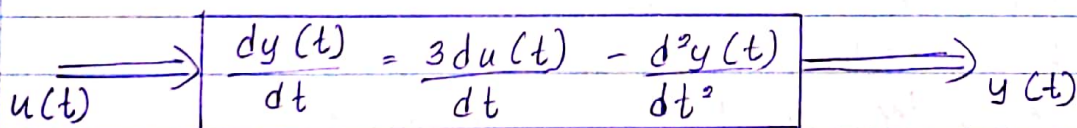
- ★ When input and output and some information about the internal dynamic of the system is known.



- ★ Easier than white box modeling.

White Box Model

- ★ When input and output and internal dynamics of the system is known.



- ★ One should know have complete knowledge of the system to derive a white box model.

Physical models (prototype)

- ★ It is a constructed copy of the modeled object.

Atlas

* Ex:- model airplane, model solar system, humanoid robot, building bridge.

Schematic models provide means of visualizing system of structure.

Organization charts

Process charts.

Graphs

Geographical Map

Diagrams.

The visual information provided by Global positioning system

Mathematical Model

Mathematical model is a representation of behaviour of real objects and phenomena in mathematical language.

Algebraic equations $\rightarrow ax^2 + bx + c$

Differential equations $\rightarrow dy/dx = Cx$

Algorithm

Formula $\rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Theorem

Ex:- (or) Marc wishes to invest in a stock with a 10% return annually. How much money will he have at the end of the year?

Marc's initial investment is unknown.

Marc's initial investment is unknown

Let us define,

* x : the amount that Marc invests in this space.

The amount accumulated at the end of the year will be,

$$= x + \frac{10}{100} x$$

$$= x + 0.1x$$

$$= 1.1x$$

This x is a initial invest.

(02) A carpenter produces and sells his own furniture. Pine tables are sold for 650\$, cherry table for 750\$, maple for 850\$. What is the carpenter's annual revenue?

Pine	Cherry	Maple
x	y	z

$$\left. \begin{array}{l} \text{Pine} - 650\$ \\ \text{Cherry} - 750\$ \\ \text{Maple} - 850\$ \end{array} \right\} 650x + 750y + 850z$$

- ★ Each pine table produce a revenue of 650\$
- ★ If x pine table are sold a revenue of 650\$ times will be obtained.
- ★ The same argument applied to the other types of tables. Consequently.

$$\underline{\text{Total revenue} = 650x + 750y + 850z}$$

(03) The three phases of a project must be undertaken sequentially, which means that one phase cannot begin before the previous phase is finished. We know that the cost of each of the phases breaks down into a fixed cost, independent of its duration, and a variable cost, which depends on the duration. The following table summarizes the situation.

Phase	1	2	3
fixed cost	318 000/-	212 000	220 000
variable cost	15 000\$/day	145 000\$/day	160 000\$/day

The designer of the project must propose a price for the project. He would like to set a price that ensures a profit margin of at least 10%. Express the total

cost of the project and the prices the designer should propose in function to the duration of each phase.

* The function of each phase is unknown.

* x = duration of phase 1 (in days)

* y = duration of phase 2 (in days)

* z = duration of phase 3 (in days)

* The cost of phase 1 can be broken down in fixed cost (318 000\$) and a variable cost (150 000\$)

* If phase 1 lasts x days, the cost of this phase will be
cost phase = 318 000 + 150 000 x

* The same principle apply to the other phase.

Total cost of project =

$$= 318\,000 + 15\,000x + 212\,000 + 14\,000y + 220\,000 + 16\,000z$$

$$= 15\,000x + 14\,000y + 16\,000z + 750\,000$$

$$= C_T + \frac{10}{100} C_T \quad (C_T - \text{Cost Total})$$

* The propose price for the project has to ensure a 10% Profit, margin for the designer.

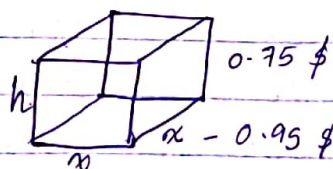
* The prime must therefore be at least 10%.

Greater than the total cost = $C_T + \frac{10}{100} C_T$

$$= 15\,000x + 14\,000y + 16\,000z + 750\,000 + \frac{10}{100} (15\,000x + 14\,000y + 16\,000z + 750\,000)$$

$$= \underline{16\,500x + 15\,400y + 16\,000z + 825\,000}$$

(04) A square bottom box without cover is made from a material that costs \$ 0.75 per square meter for the sides and \$ 0.95 per square meter for the bottom. Express the total cost of the material required to construct the box in function of its width and height.



Atlas

Let us define,

x = length of the side of the bottom of the box

h = height of the box

$$= 0.95x^2 + 4xh \times 0.75$$

$$C_T = 0.95x^2 + 3xh$$

(os) I travelled 432 km on 48 l of petrol in my car. I have to go by my car to a place which is 180 km about, How much petrol I need,

$$432 \text{ km} \rightarrow 48 \text{ l}$$

$$1 \text{ km} \rightarrow \frac{48 \text{ l}}{432 \text{ km}} \times 1 \text{ km}$$

$$\frac{48 \text{ l}}{432}$$

$$\text{Therefore, } 180 \text{ km} \rightarrow \frac{48 \text{ l} \times 180}{432}$$

$$\frac{48 \times 180}{432} \text{ l}$$

$$\underline{20 \text{ l}}$$

Formulation

Petrol need for travelling 432 km = 48 l

Petrol need for travelling 180 km = ?

Mathematical Description

* x = distance I travel

* y = petrol I need.

y varies directly x ,

$$y = kx$$

Ames

$$48 = k \cdot 432$$

$$k = \frac{48}{432} = \frac{1}{9}$$

$$y = \frac{1}{9}x$$

$$y = \frac{1}{9} \times 180 \\ = 20 \text{ km}$$

- (06) A motor boat goes upstream on a river and covers the distance between two towns on the riverbank in six hours. It covers this distance down stream in five hours. If the speed of the stream is 2 km/h. Find the speed of the boat in still water.

Formulation

We know the speed of the river and the time taken to cover the distance between two places. We have to find the speed of boat in still water.

Mathematical Description

Let x for the speed of the boat, t for the time taken and y for the distance travelled.

Then $x = y/t$

$$y = xt$$

$x \rightarrow \text{speed}$

$$\leftarrow x-2 = d/6 \text{ --- (1)}$$

$t \rightarrow \text{time}$

$$\rightarrow x+2 = d/5 \text{ --- (2)}$$

$y \rightarrow \text{distance}$

from (1), $d = 6(x-2) \text{ --- (3)}$

from (2), $d = 5(x+2) \text{ --- (4)}$

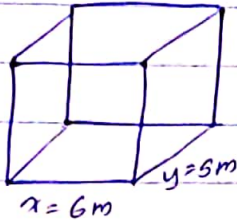
$$(3) = (4)$$

$$6(x-2) = 5(x+2)$$

$$6x - 12 = 5x + 10$$

$$x = 22 \text{ km/h}$$

- (02) Suppose you have a room of length 6m and width 5m. You execute to cover the floor of the room with square mosaic tiles of side 30cm. How many tiles will you need? Solve this by constructing a mathematical model?



length of the room = x

width of the room = y

length of one tile = z

area of tile = z^2

area of the room = xy

number of the tile = k

$$k = \frac{xy}{z^2}$$

$k = \frac{\text{Area of room}}{\text{Area of one tile}}$

$= \frac{6 \times 5}{9 \times 10^{-2}}$

$$= \frac{30}{9} \times 10^2 = 333.$$

$$= \frac{30}{9} \times 10^2 = 333.$$

Modelling with first order Differential Equations.

Find order and degree of following differential equations.

(01) $\frac{d^3y}{dx^3} + \frac{d^2y}{dy^2} + 2y \frac{dy}{dx} + xy = 0 \rightarrow$ order three and degree 1

(02) $3 \frac{dy}{dx} + x = \left(y^2 - x \left(\frac{dy}{dx} \right)^4 \right) \rightarrow$ order one and degree 5

(03) $\frac{d^3y}{dx^3} - x \frac{dy}{dx} + 3 = 0 \rightarrow$ order one and degree and degree does not exists.

Example 01

Show that equation $dy/dx = 2x$, $y(0) = 0, y(1) = 100$ has no solution.

$$\frac{dy}{dx} = 2x$$

$$\textcircled{1} \Rightarrow 0 = 0 + C \Rightarrow C = 0$$

$$y(1) = 100$$

$$dy = 2x dx$$

$$x = 1, y = 100$$

$$\int dy = \int 2x dx$$

$$y = x^2 + C$$

$$y = 2x^2 + C$$

$$100 = 1 + C$$

$$y = x^2 + C \text{ --- } \textcircled{1}$$

$$99 = C$$

$$y(0) = 0$$

$$\Rightarrow x = 0, y = 0$$

Linear and non-linear ordinary Differential Equation.

An ordinary differential equations of order n is said to be a linear equations if it is of the form.

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y = f(x)$$

When $a_0(x), a_1(x), \dots, a_n(x)$ and $f(x)$ are given functions that defined on an interval and $a_0(x)$ is not identically zero. An equation that is not linear is said to be a non equations.

For example;

$$\frac{dy}{dx} + (\cos x)y = e^x \rightarrow \text{linear differential equation}$$

$$x \frac{d^3 y}{dx^3} - e^x \frac{dy}{dx} + \sin xy = 0 \rightarrow \text{linear differential equation}$$

$$\frac{dy}{dx} + y^2 = 1 \rightarrow \text{non linear equation}$$

$$\frac{d^2y}{dx^2} + \cos(xy) \frac{dy}{dx} = \sin x \rightarrow \text{Non linear differential equation}$$

$$\frac{d^3y}{dx^3} - x \left(\frac{dy}{dx} \right)^2 + y = 0 \rightarrow \text{Non linear differential equation}$$

Find the order of each of the following differential equations and determine whether it is linear or non linear.

$$\frac{d^2u}{dx^2} + xy \frac{du}{dx} + y = 2 \rightarrow \text{non linear differential equation}$$

$$y \frac{d^3y}{dx^3} + \frac{dy}{dx} = 0 \rightarrow \text{non linear differential equation}$$

$$\frac{d^2y}{dx^2} + xy = \sin x \rightarrow \text{linear differential equation}$$

$$\frac{d^2y}{dx^2} + 4 \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 3y = \sin x \rightarrow \text{linear differential equation}$$

Separable Differential Equation

$$(o) (x+1) \frac{dy}{dx} = 2y$$

$$(x+1) dy = 2y dx$$

$$\frac{1}{2y} dy = \frac{1}{(x+1)} dx$$

By intergrating both sides,

$$\int \frac{1}{2y} dy = \int \frac{1}{x+1} dx$$

$$\frac{1}{2} \int \frac{1}{y} dy = \int \frac{1}{x+1} dx$$

$$\frac{1}{2} \ln y = \ln |x+1| + C$$

$$\ln y^{1/2} = \ln |x+1| + C$$

$$y^{1/2} = x+1 + C$$

where C is arbitrary constant.

Atlas

First Order Linear Equations.

The simplest 1st order equations are those that can be written in the form.

$$\frac{dy}{dx} = g(x)$$

For example

$$\frac{dy}{dx} = x^3$$

$$dy = x^3 dx$$

$$\int dy = \int x^3 dx$$

$$y = x^4/4 + c$$

where c is arbitrary constant.

This is a general solution.

More generally 1st order differential equation is,

$$\frac{dy}{dx} + P(x)y = Q(x)$$

For example

$$\frac{dy}{dx} + x^2 y = e^x$$

$$\frac{dy}{dx} + 5y = 2$$

$$P(x) = x^2 \quad Q(x) = e^x$$

$$P(x) = 5, \quad Q(x) = 2$$

$$\frac{dy}{dx} + (\sin x)y + x^3 = 0$$

$$P(x) = \sin x, \quad Q(x) = x^3$$

Integrating factor

$$m \text{ or } M = e^{\int P(x) dx}$$

$$\frac{d \ln}{dx} = \frac{d}{dx} (e^{\int P(x) dx})$$

$$= e^{\int P(x) dx} \frac{d \int P(x) dx}{dx}$$

$$\frac{d(M)}{dx} = MP(x)$$

$$dx$$

$$\frac{d(My)}{dx} = M \frac{dy}{dx} + y \frac{dM}{dx}$$

$$\frac{d(My)}{dx} = M \frac{dy}{dx} + y MP(x)$$

example

$$\frac{dy}{dx} - y = e^{2x} \text{ --- (1)}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$I.F = m = e^{\int P(x) dx}$$

$$m = e^{-\int dx} = e^{-x}$$

$$(1) \times m$$

$$\frac{dy}{dx} \cdot e^{-x} - y \cdot e^{-x} = e^{2x} \cdot e^{-x}$$

$$\frac{d(ye^{-x})}{dx} = e^x$$

$$d(ye^{-x}) = e^x dx$$

Integrating are both side,

$$\int d(ye^{-x}) = \int e^x dx$$

$$ye^{-x} = e^x + c$$

$$y = e^{2x} + ce^x$$

where c is an arbitrary constant.

Solve the initial value problem (IVP)

$$x \frac{dy}{dx} - y = x \quad y(1) = 2$$

$$x \frac{dy}{dx} - y = x$$

$$\frac{dy}{dx} - \frac{y}{x} = 1 \text{ --- (1)}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\begin{aligned} m = I.F &= e^{\int -1/x dx} \\ &= e^{-\ln x + c} \\ &= e^{-\ln x} \end{aligned}$$

$$m = I.F = e^{\ln x^{-1}} e^c$$

Atlas

$$e^{\ln x^{-1}} = x^{-1} = 1/x$$

$$m = I.F = 1/x \cdot e^c$$

① $\times m$

$$\frac{e^c}{x} \left(\frac{dy}{dx} \right) - \frac{y}{x} \frac{e^c}{x} = \frac{e^c}{x}$$

$$\frac{d(ye^{c/x})}{dx} = \frac{e^c}{x}$$

$$d(ye^{c/x}) = (e^c/x) dx$$

By integrating both sides,

$$\int d(ye^{c/x}) = \int e^c/x dx$$

$$ye^{c/x} = e^c \int 1/x dx$$

$$y/x = \ln x + c$$

where c is an arbitrary constant.

$$y(1) = 2$$

$$x=1, y=2$$

$$y/x = \ln x + c$$

$$2/1 = 0 + c$$

$$c = 2$$

$$\therefore y = x \ln x + 2x$$

Solving 2nd Order Differential Equation Using I.F

$$y'' + P(x)y' = Q(x) \text{ --- (1)}$$

Substitute, $y' = u$

$$u' + P(x)u = Q(x) \text{ --- (2)}$$

Next Calculation,

$$M = e^{\int P(x) dx}$$

② $\times M$

$$Mu' + MuP(x) = Q(x) \cdot M$$

$$\frac{d(Mu)}{dx} = M Q(x)$$

$$d(Mu) = M Q(x) dx$$

Integrating both sides,

$$\int d(Mu) = \int M Q(x) dx + c$$

$$\underline{Mu = \int M Q(x) dx + c}$$

$$(01) \frac{dy}{dx} - \frac{3y}{x+1} = (x+1)^4 \quad \text{--- (1)}$$

$$M = e^{\int P(x) dx}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$M = e^{\int -(3/x+1) dx}$$

$$M = e^{-3 \int 1/x+1 dx}$$

$$M = e^{-3} \ln |x+1| + C$$

$$M = e^{\ln (x+1)^{-3}} e^C$$

because of $e^C = k$

$$M = e^{\ln (x+1)^{-3}} \cdot k \quad \text{--- (2)}$$

$$e^{\ln (x+1)^{-3}} = (x+1)^{-3}$$

$$[k \cdot (x+1)^{-3}] \frac{dy}{dx} - k(x+1)^{-3} \left[\frac{3y}{x+1} \right]$$

$$= (x+1)^4 \times 3y/x \times (x+1)^4 \times k(x+1)^{-3}$$

$$d [k (x+1)^{-3} y] = k (x+1)$$

$$d [k (x+1)^{-3} y] = k (x+1) dx$$

By integrating both sides,
 $\int d [k (x+1)^{-3} y] = \int k (x+1) dx$

$$\int d [k (x+1)^{-3} y] = \int k (x+1) dx$$

$$k (x+1)^{-3} y = k \int (x+1) dx$$

$$y = \int x dx + \int dx$$

$$\frac{y}{(x+1)^3} = \frac{x^2}{2} + x + C'$$

$$y = \frac{1}{2} x^2 (x+1)^3 + x (x+1)^3 + C' (x+1)^3$$

where C' is an arbitrary constant.

2nd order differential equation.Eqⁿ types.

$$\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Ey = 0$$

Order the order is the highest derivative.

Example,

$$\frac{dy}{dx} + y^2 = 3x$$

$$\frac{d^3y}{dx^3} + x \frac{dy}{dx} + y = e^x$$

1st derivative and 1st order3rd derivative and 3rd order.

$$\frac{d^2y}{dx^2} + xy = \sin x$$

2nd derivative and 2nd order

$$\frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = f(x)$$

where $P(x)$, $Q(x)$ and $f(x)$ are function of x Case I:- if $f(x) = 0$ then it is called homogeneous equation

Example,

$$\text{Solve } \frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0 \text{ using } y = e^{rx}$$

$$r^2 e^{rx} + r e^{rx} - 6 e^{rx} = 0$$

$$e^{rx} (r^2 + r - 6) = 0$$

$$e^{rx} \neq 0$$

$$\text{then } r^2 + r - 6 = 0 \leftarrow \text{characteristic eq.}$$

$$r^2 + r - 6 = 0$$

$$(r-2)(r+3) = 0$$

$$r=2 \quad \text{or} \quad r=-3$$

And we have two solution.

$$y = e^{2x} \quad \& \quad y = e^{-3x}$$

but general solution - $y = Ae^{2x} + Be^{-3x}$

Check,

$$y = Ae^{2x} + Be^{-3x}$$

$$\frac{dy}{dx} = 2Ae^{2x} - 3Be^{-3x}$$

$$\frac{d^2y}{dx^2} = 2Ae^{2x} + 9Be^{-3x}$$

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

$$L.H.S \Rightarrow \frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y$$

$$= Ae^{2x} + Be^{-3x} + 2Ae^{2x} - 3Be^{-3x} - 6(Ae^{2x} + Be^{-3x})$$

$$= 3Ae^{2x} - 2Be^{-3x} - 12Ae^{2x} - 54Be^{-3x}$$

$$= -(9Ae^{2x} + 56Be^{-3x})$$

$$\text{Solve } \frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 20y = 0 \quad \text{using } y = e^{rx}$$

$$r^2e^{rx} - 9re^{rx} + 20e^{rx} = 0$$

$$e^{rx}(r^2 - 9r + 20) = 0$$

$$e^{rx} \neq 0$$

$$\text{then } r^2 - 9r + 20 = 0$$

$$(r-4)(r-5) = 0$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{9 \pm \sqrt{81 - 4 \times 1 \times 20}}{2}$$

$$= \frac{9 \pm \sqrt{81 - 80}}{2}$$

$$= \frac{9 \pm \sqrt{1}}{2}$$

$$r = 4 \quad \text{or} \quad r = 5$$

And we have two solution

$$y = e^{4x} \quad \& \quad y = e^{5x}$$

$$\text{but general solution } y = Ae^{4x} + Be^{5x}$$

Complex roots

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$$

$$y = e^{rx}$$

$$\frac{dy}{dx} = re^{rx}$$

$$r^2 e^{rx} - 4r e^{rx} + 13 e^{rx} = 0$$

$$e^{rx} (r^2 - 4r + 13) = 0$$

$$\frac{d^2y}{dx^2} = r^2 e^{rx}$$

$$e^{rx} \neq 0$$

$$\text{the } r^2 - 4r + 13 = 0$$

$$r = \frac{4 \pm \sqrt{16 + 4 \times 1 \times 13}}{2}$$

$$r = 2 + \sqrt{17} \quad \text{or} \quad 2 - \sqrt{17}$$

$$= \frac{4 \pm \sqrt{16 + 52}}{2}$$

$$y = A e^{2+}$$

$$= \frac{4 \pm \sqrt{68}}{2}$$

$$= \frac{4 \pm \sqrt{4 \times 17}}{2}$$

$$= \frac{4 \pm 2\sqrt{17}}{2}$$

$$= \frac{2 \pm \sqrt{17}}{1}$$

H.W/ (01) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 6y = 0$

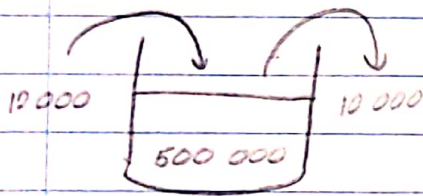
(02) $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$

(03) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = 0$

Modelling with 1st order differential equation

example,

A pond initially contains 500 000 gallons of unpolluted water has an outlet that reduces 10 000 gallons of water per day. A stream flows in to the pond at 12 000 gallons per day, containing water with a concentration of 2 grams per gallon of a pollutant. Find a differential equation that models this process and determine what the concentration of pollutant will be after 10 days.



2 grams / gallon pollutant

we let $x(t)$ be amount of pollutant in grams in the pond after t days.

We use the fundamental property rate

Total rate = Rate in - Rate out

To find the rate in we use

$$\frac{\text{grams}}{\text{day}} = \frac{\text{gallons}}{\text{day}} \times \frac{\text{grams}}{\text{gallons}}$$

$$= 12\,000 \times 2$$

$$= 24\,000 \text{ grams per day.}$$

To find the rate out as 1st notice that since there was initially 500 000 gallons of water in the pond and the water remove

After

is increasing at a rate of 12 000 gallons per day, the total number of gallons of water in the level, after t day is gallons,

$$= 500\,000 + 2000t$$

The unit per the rate out is grams per day, we write

$$\frac{\text{grams}}{\text{day}} = \frac{\text{gallons}}{\text{day}} \times \frac{\text{grams}}{\text{gallon}}$$

$$= \frac{12000}{1} \times \frac{x}{(500\,000 + 2000t)}$$

$$= \frac{10x}{500+2t}$$

Putting this all together

$$\frac{dx}{dt} - 24\,000 = \frac{10x}{500+2t}$$

let $x(t)$ be a amount of pollutant in grams in the pond after t days.

$$\frac{dx}{dt} + \frac{10x}{500+2t} = 24\,000$$

$$\begin{aligned} I.F &= e^{\int \frac{10}{500+2t} dt} \\ &= e^{5 \ln(500+2t)} \\ &= e^{\ln(500+2t)^5} \\ &= (500+2t)^5 \end{aligned}$$

$$\frac{d}{dt} (x (500+2t)^5) = 24\,000 (500+2t)^5$$

$$x (500+2t)^5 = 24\,000 \frac{(500+2t)^6}{2 \times 6} + C$$

$$x = 200 (500 + 200 + 2000(500 + 2t)) + \frac{C}{(500+2t)^5}$$

$$t=0, \quad x=0$$

$$0 = 2000(500)^6 + \frac{C}{(500)^5}$$

$$C = -3.125 \times 10^{19}$$

$$x = 2000 (500 + 2t) + \frac{-3.125 \times 10^{19}}{(500 + 2t)^5}$$

$$t = 10$$

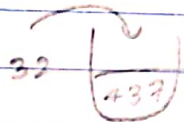
$$x = 2000 (500 + 20) + \frac{-3.125 \times 10^{19}}{(500 + 20)^5}$$

$$= 216\ 072\ \text{grams} //$$

Growth Model

All around us, the population of people, animals and object are expanding. we can better prepare for the future by understanding how things develop. In this part concentrate on ~~two~~ time-dependent change.

Example/ Marco is a collection of antique soda bottles. His collection currently contains 437 bottles. Every years he budgets enough money to buy 32 new bottles. Can we determine how many bottles he will have in 5 years, and how long it will take for his collection to reach 1000 bottles.



$$P_0 = 437$$

$$P_n = P_{n-1} + 32$$

This is called recursive relationship.

A recursive relationship is a formula which relates the next value in a sequence to the previous values.

Here the number of bottles in year n can be found by adding 32 to the number of bottles in previous years.

P_{n-1} using this relationship;

$$P_1 = P_0 + 32 = 437 + 32 = 469$$

$$P_2 = P_1 + 32 = 469 + 32 = 501$$

$$P_3 = P_2 + 32 = 501 + 32 = 533$$

$$P_4 = P_3 + 32 = 533 + 32 = 565$$

$$P_5 = P_4 + 32 = 565 + 32 = 597$$

$$P_n = P_0 + n(32)$$

$$\begin{aligned}
 P_3 &= 237 + 5 \times 37 \\
 &= 237 + 185 \\
 &= 402 \quad 422 \quad 597
 \end{aligned}$$

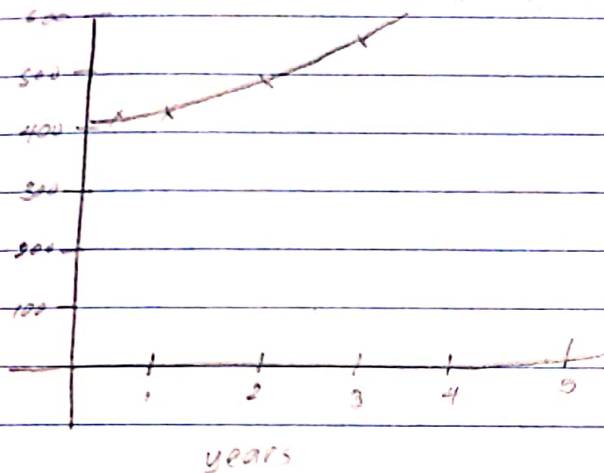
how long it will take this collection for reach 1000.

$$1000 = 237 + n(37)$$

$$n = 17.57 \rightarrow n \approx 18$$

nearly 18 years.

Moncc collection grew by the same number of bottles every year, we can see the time the shape of timer growth



Defⁿ

Linear growth

If a quantity starts at side P_0 and grows by A , every time period, then the quantitative after n time periods can be determined as follows,

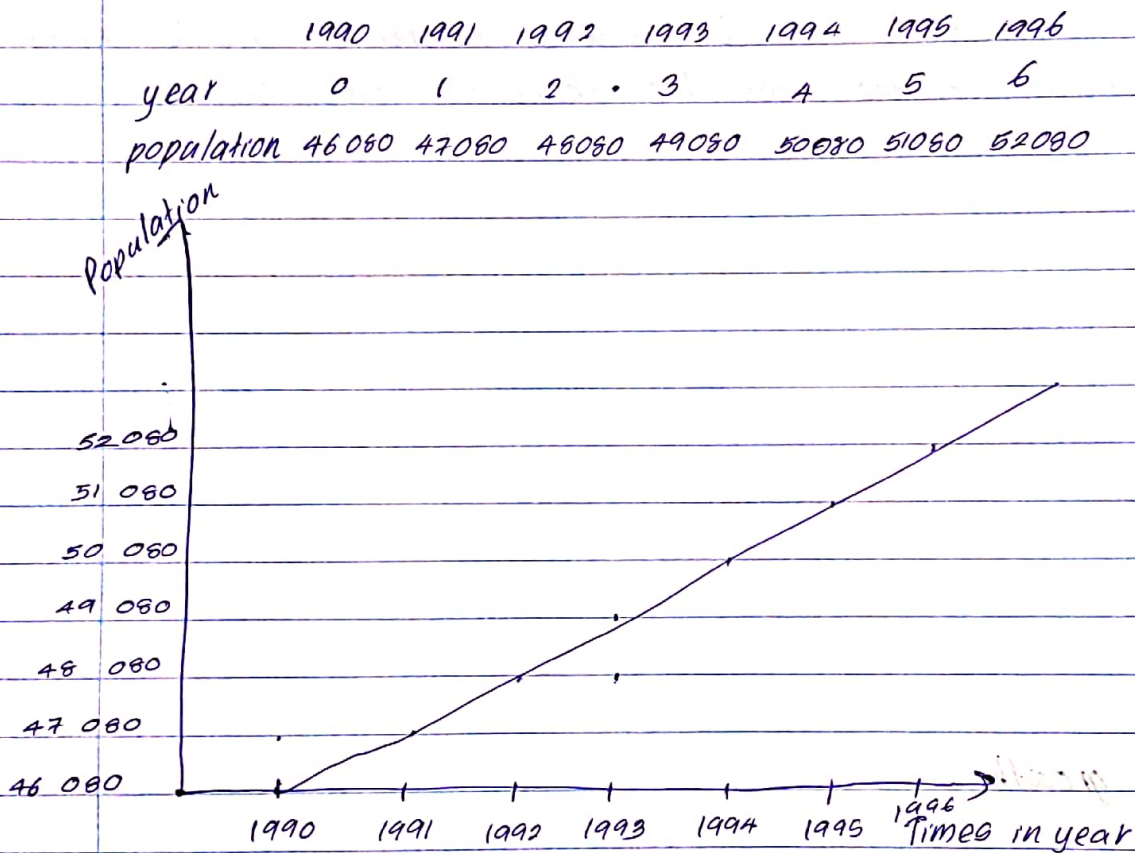
$$P_n = P_0 + nA$$

H.W. The population of _____ in a national forest was measured to be 12 000 in 2003. and

Linear growth

A quantity grows linearly if it grows by a constant amount for each unit of time.

Ex:- Suppose in Arizona, the number of residents ^{increased} by 1000 people per day. If the initial population was 46080 in 1990, Can you predict the population in 2023.



This is the graph of the population growth over the six period in Arizona. It is a straight line and can be modeled with a linear growth model.

The initial population P_0 is 46080

The future population on the number of years, k , after the initial year.

$$P = P_0 + 1000k \Rightarrow P = 46080 + 1000k$$

Atlas

We predict the population 2023

1990 $\rightarrow$ 2023

$$k = 33$$

$$P = 46.060 + 1000 \times 33$$

$$= 46\,060 + 33\,000$$

$$= 79\,060$$

Linear Growth Model

Linear growth begins with initial population called P_0 . In each time period or generation L , the population changes by a constant amount called the common difference d . The basic model is $P(L) = P_0 + Ld$.

ex:- Dora has inherited a collection of 30 antique frogs. Each year she vows to buy two frog's a month to grow the collection. This is an additional 24 frogs per year. How many frogs will she have in six years. How long will it take her to reach 510 frogs.

$$\begin{aligned} P(L) &= P_0 + Ld \\ &= 30 + 24 \times 6 \\ &= 30 + 144 \\ &= 174 \end{aligned}$$

$$\begin{aligned} P(L) &= P_0 + Ld \\ 510 &= 30 + 2 \times d_{\text{months}} \quad 24 \times d \\ d_{\text{months}} &= \frac{480}{2} \end{aligned}$$

$$d_{\text{months}} = 240.$$

10 years will take to reach 510.

Assume a car depreciates by the same amount each year. Toe purchase a car in 2010 for \$ 16,800. ^{In 2014} it is worth \$ 12000. Find the linear growth model. Predict how much car will be worth in 2023

$$\begin{aligned} d &= \frac{16\,800 - 12\,000}{4} \\ &= \frac{4\,800}{4} \\ &= 1\,200 \end{aligned}$$

$$\begin{aligned} P(L) &= P_0 + Ld \\ &= 16\,800 + (-1200) \times 13 \\ &= 16\,800 - 15\,600 \\ \text{Atlas} &= \$ 1\,200 \end{aligned}$$

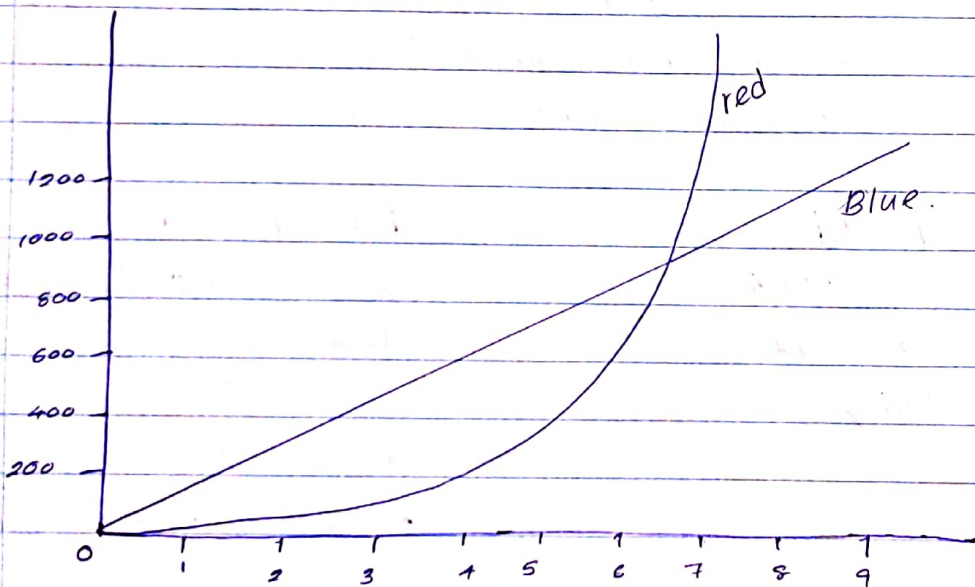
Exponential Growth

Exponential growth happens when an initial population increase by the same percentage or factor over equal time increment or generations. This is known as relative growth and it usefully expressed as percentage.

ex:- Let's say population is growing by 1.6% each year. For every 1000 people in the population; there will be $1000 \times 0.016 = 16$ more people added for year.

A quality grows exponentially if it grows by a constant factor or rate for each unit of time.

Graphical comparison of Linear and Exponential Growth.



In this graph

ex:- A city is growing at a rate of 1.6% per year. The initial population in 2010 is $P_0 = 125\,000$. Calculate the city's population over the next few years.

growth rate 1.6%

$$P_0 = 125\,000 \quad (\text{In } 2010)$$

$$\begin{aligned} \text{Population in } 2011 &= 125\,000 + 1.6 \times 125\,000 \\ &= 125\,000 + \frac{100}{100} \times 1250 \\ &= 125\,000 + 2000 \\ &= 127\,000 \end{aligned}$$

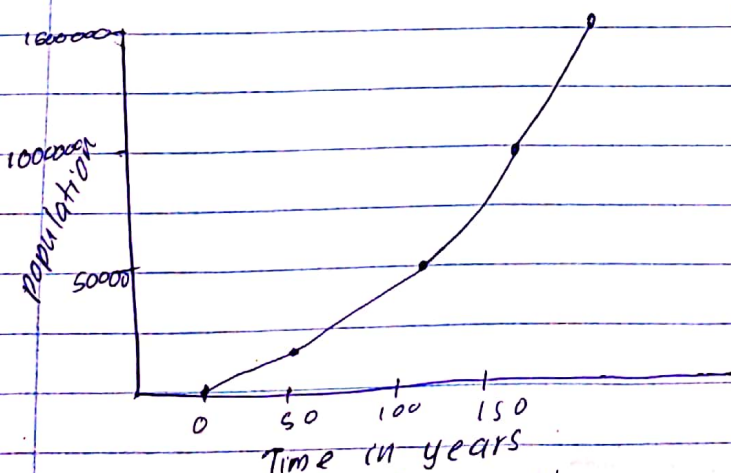
$$\begin{aligned} \text{Population in } 2012 &= 127\,000 \times 1.016 \\ &= 129,032 \end{aligned}$$

$$\begin{aligned} \text{" " } 2013 &= 129,032 \times 1.016 \\ &= 131,197 \end{aligned}$$

$$\begin{aligned} \text{" " } 2014 &= 125\,000 \times (1.016)^4 \\ &= 133,194 \end{aligned}$$

$$\text{model of this ; } P(t) = 125\,000 (1 + 0.016)^t$$

St. Louis, Missouri has declined in population at a rate of 1.6% per year over the last 60 years. The population in 1950 was 857 000. Find the population in 2024.



Graph of city growth

Each year the population is 1.6% more than the previous year

$$P(t) = P_0 (1 + r)^t$$

The graph $P_0 = 857\,000$

Atlas

$$\begin{aligned}
 \text{Population } 1951 &= 657\,000 - 657\,000 \times 0.016 \\
 &= 657\,000 - 10\,512 \\
 &= 646\,488
 \end{aligned}$$

The relative growth rate is 1.6%. This means 1.6% of the population is subtracted from 100% of the population that already exists each year. This is a factor of $(1 - 0.016) = (0.984)$.

$$\begin{aligned}
 1952 &= 646\,488 (0.984) \\
 &= 636\,795
 \end{aligned}$$

$$\begin{aligned}
 1953 &= 646\,488 (0.984) \\
 &= 636\,795
 \end{aligned}$$

Now can create an equation for the city's growth. Each year population is 1.6% less than the previous year.

$$P(t) = 657\,000 (1 - 0.016)^t$$

$$t = 74$$

$$\begin{aligned}
 P(74) &= 657\,000 (1 - 0.016)^{74} \\
 &= 657\,000 \times 0.3031 \\
 &= 199\,156.41
 \end{aligned}$$



Note/ The graph of the population of St. Lours, mission over time follows a declining exponential growth model.

Atlas

Exponential growth model

$$P(t) = P_0 (1+r)^t$$

P_0 is the initial population.

t is the time unit

r is the relative growth rate

r is positive if the population is increasing and negative when the population is decreasing.

The average inflation rate of the U.S. dollar over the last five years is 1.7% per year. If a new car cost \$18 000 five years ago how much would it cost today.

$$P(t) = P_0 (1+r)^t$$

$$= 18\,000 (1+0.017)^t$$

$$t = 5$$

$$P(5) = 18\,000 (1+0.017)^5$$

$$= 19\,582.91 \$$$

In May of 2022 there were 15 cases of Ebola in Sierra Leone. By August, there were 850 cases. If the virus is spreading at the same rate (exponential growth), how many cases will there be in February of 2023.

$$P(t) = P_0 (1+r)^t$$

$$850 = 15 (1+r)^3$$

$$(1+r)^3 = 56.66$$

$$1+r = 3.84$$

$$r = 2.84$$

$$P(t) = P_0 (1+r)^t$$

$$P(6) = 850 (3.84)^6$$

$$= 2\,725\,250$$

No: _____

Date: ____/____/____

According to a new forecast, the population of Puerto Rico is in decline. If the population in 2010 is 3 978 000 and the prediction for the population in 2050 is 3, 697 000. find the annual percent decrease rate.

Difference Equations.

Differential eq^{ns} are applicable systems and cannot be used for discrete variable. Difference equations are the discrete equivalent of differential equation and arise whenever, an independent variable can have only discrete value.

Some useful result

' Δ ' is a forward difference operator such that $\Delta f(x) = f(x+h) - f(x)$

$$\Delta y_x = y_{x+1} - y_x$$

$$\Delta y_0 = y_1 - y_0$$

$$\Delta y_1 = y_2 - y_1$$

$$\vdots$$

$$\Delta y_n = y_{n+1} - y_n$$

$$\begin{aligned} \text{Also } \Delta^2 y_0 &= \Delta(\Delta y_0) \\ &= \Delta(y_1 - y_0) \\ &= \Delta y_1 - \Delta y_0 \\ &= y_2 - y_1 - (y_1 - y_0) \\ &= y_2 - 2y_1 + y_0 \end{aligned}$$

$$\begin{aligned} \Delta^3 y_0 &= \Delta(\Delta^2 y_0) \\ &= \Delta(y_2 - 2y_1 + y_0) \\ &= \Delta y_2 - 2\Delta y_1 + \Delta y_0 \\ &= y_3 - y_2 - 2(y_2 - y_1) + y_1 - y_0 \\ &= y_3 - y_2 - 2y_2 + 2y_1 + y_1 - y_0 \\ &= y_3 - 3y_2 + 3y_1 - y_0 \end{aligned}$$

$$\Delta^n y_0 = y_n - nC_1 y_{n-1} + nC_2 y_{n-2} - \dots + (-1)^{n-1} nC_{n-1} y_1 + (-1)^n y_0$$

Generally,

$$\Delta^n y_r = y_{n+r} - nC_1 y_{n+r-1} + nC_2 y_{n+r-2} - \dots + (-1)^r y_0$$

Also

Example/

Evaluate the following.

i) Δe^x ii) $\Delta^2 e^x$ iii) $\Delta \tan^{-1} x$ iv) $\Delta \left(\frac{x+1}{x^2-3x+2} \right)$

$$\Delta e^x = e^{x+h} - e^x = e^x (e^h - 1)$$

$$\Delta e^x = e^x (e - 1) \quad \text{if } h = 1$$

$$\begin{aligned} \text{ii) } \Delta^2 e^x &= \Delta (\Delta e^x) \\ &= \Delta (e^x (e - 1)) \\ &= (e - 1) \Delta e^x \\ &= (e - 1) e^x (e - 1) \\ &= e^x (e - 1)^2 \end{aligned}$$

$$\begin{aligned} \text{iii) } \Delta \tan^{-1} x &= \tan^{-1}(x+h) - \tan^{-1} x \\ &= \tan^{-1} \left(\frac{x+h-x}{1+(x+h)x} \right) \\ &= \tan^{-1} \left(\frac{h}{1+(x+h)x} \right) \end{aligned}$$

(Note: The original image has a scribble over the final expression, which appears to be $\Delta \left[\frac{x+1}{x^2-3x+2} \right]$)

$$\begin{aligned} \text{iv) } \Delta \left[\frac{x+1}{x^2-3x+2} \right] &= \Delta \left[\frac{(x+1)}{(x-1)(x-2)} \right] \\ &= \Delta \left[\frac{A}{x-1} + \frac{B}{x-2} \right] \end{aligned}$$

$$\frac{x+1}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$$

$$x+1 = A(x-2) + B(x-1)$$

$$1 = A + B$$

$$-2 = -2A - B$$

$$A = -2$$

$$B = 3$$

$$\begin{aligned} &= \Delta \left[\frac{-2}{x-1} + \frac{3}{x-2} \right] \\ &= -2 \Delta \left(\frac{1}{x-1} \right) + 3 \Delta \left(\frac{1}{x-2} \right) \\ &= -2 \left[\frac{1}{x+1-1} - \frac{1}{x-1} \right] + 3 \left[\frac{1}{x+1-2} - \frac{1}{x-2} \right] \\ &= \frac{-(x+4)}{x(x-1)(x-2)} \end{aligned}$$

Also

The shift operator 'E' is defined as $E(f(x)) = f(x+h)$
 $E^2(f(x)) = E[E(f(x))]$
 $= E[f(x+h)]$
 $= f(x+2h)$

$$E^n(f(x)) = f(x+nh)$$

Moreover $E^{-1}(f(x)) = f(x-h)$; where E^{-1} is the inverse operator.

Relation between Δ and E.

Relation between Δ and E is given by $E = 1 + \Delta$

Proof/ $\Delta y_n = y_{n+1} - y_n$

but $E(y_n) = y_{n+1}$

$$\therefore \Delta y_n = E y_n - y_n$$

$$\Delta y_n = (E - E^0) y_n$$

$$\Delta = (E - 1)$$

$$E = \Delta + 1$$

Factorial Notation of a Polynomial.

A product of the form $x(x-1)(x-2) \dots (x-r+1)$ is called a factorial and is denoted by $[x]^r$.

$$[x] = x$$

$$[x]^2 = x(x-1)$$

$$[x]^3 = x(x-1)(x-2)$$

$$\vdots$$

$$[x]^n = x(x-1)(x-2) \dots (x-(n-1))$$

$$\Delta [x]^n$$

$$\Delta[x] = x+1 - x = 1$$

$$\Delta[x]^2 = 2(x+1)(x+1-1) - 2x(x-1)$$

$$= 2x(x+1) - 2x(x-1)$$

$$= 2x^2 + 2x - 2x^2 + 2x$$

$$= 0$$

$$\Delta[x]^3 = 3(x+1)(x+1-1)(x+1-2) - 3x(x-1)(x-2)$$

$$= 3(x+1)(x-1)x - 3x(x-1)(x-2)$$

$$= 3x(x-1)[x+1 - x+2]$$

$$= 9x(x-1)$$

Note,

- 1) Every polynomial of degrees can be expressed as a factorial polynomial of the same degree and vice versa.
- 2) The coefficient of highest power of x and also the constant term remains unchanged while transforming a polynomial equation to factorial notation.

example 2.

Express the polyⁿ $2x^2 - 3x + 1$ in factorial notation,

$$[x] = x$$

$$[x]^2 = 2x(x-1)$$

$$2x^2 - 3x + 1 = 2x^2 - 2x - x + 1$$

$$= 2x(x-1) - x + 1$$

$$= [x]^2 - [x] + 1$$

Express the polyⁿ $3x^2 - x + 2$ in factorial notation.

$$[x] = x$$

$$[x]^2 = 2x(x-1)$$

$$[x]^3 = 3x(x-1)(x-2)$$

$$3x^2 - x + 2 \rightarrow 2x^2 + x^2 - x + 2 \quad 3x^2 - x + 2 = 3[x]^3 +$$

$$= 2x(x-1) + x^2 + 2 + x$$

$$3x^3 - x + 2 = 3[x]^3 + A[x]^2 + B[x] + 2$$

$$A = 0, B = 0$$

$$= 9x(x-1)(x-2) + A2x(x-1) + Bx + 2$$

$$= 9x(x^2 - 3x + 2) + 2Ax(x-1) + Bx + 2$$

Order of differential eqⁿ

Order of differential eqⁿ is the difference between the largest and the smallest argument occurrence in the differential equation

$$\Rightarrow x_{n+1} - x_n = 2 \quad \text{order} = (n+1) - n = 1$$

$$\Rightarrow x_{n+2} - x_{n+1} + 2x_n = n \quad \text{order} = n+2 - n = 2$$

In general, the higher the order of an eqⁿ, the more difficult it is solve.

Example/

Write the given difference eqⁿ in the subscript notation

$$\Delta^3 y_x + \Delta^2 y_x + 2y_x + y_x = 0$$

$$\Delta y_x = y_{x+1} - y_x$$

$$\Delta^2 y_x = \Delta(\Delta y_x)$$

$$= \Delta(y_{x+1} - y_x)$$

$$= \Delta y_{x+1} - \Delta y_x$$

$$= y_{x+2} - y_{x+1} - y_{x+1} + y_x$$

$$= y_{x+2} - 2y_{x+1} + y_x$$

$$\Delta^3 y_x = \Delta(\Delta^2 y_x)$$

$$= \Delta(y_{x+2} - 2y_{x+1} + y_x) = \Delta y_{x+2} - 2\Delta y_{x+1} + \Delta y_x$$

$$= y_{x+3} - y_{x+2} - 2(y_{x+2} - y_{x+1}) + y_{x+1} - y_x$$

$$= y_{x+3} - 3y_{x+2} + 4y_{x+1} - y_x$$

$$\begin{aligned} & \Delta^3 y_x + \Delta^2 y_x + \Delta y_x + y_x \\ &= y_{x+3} - 3y_{x+2} + 3y_{x+1} - y_x + y_{x+1} - 2y_{x-1} + y_x + y_{x+1} - y_x + y_x \\ &= y_{x+3} + 2y_{x+2} + 2y_{x+1} \end{aligned}$$

Find the differential equation satisfied by the relation
 $y = A2^n + n3^{n-1}$. Given that $y_n = A2^n + n3^{n-1}$ — (1)
 we only need 1st difference

$$y_{n+1} = A2^{n+1} + (n+1)3^{n+1-1}$$

$$y_{n+1} = A2^{n+1} + (n+1)3^n \text{ — (2)}$$

$$(2) - (1)$$

$$y_{n+1} - 2y_n = A2^{n+1} + (n+1)3^n - 2A2^n - 2n3^{n-1}$$

$$y_{n+1} - 2y_n = A2^{n+1} + n3^n + 3^n - 2A2^n - 2n3^{n-1}$$

$$= (n+3)3^{n-1}$$

Find the difference eqⁿ satisfied by the relation
 $y = ax^2 - bx$. Given that $y_x = ax^2 - bx$ — (1)
 taking the 1st difference

$$\Delta y_x = a \Delta(x^2) - b \Delta x$$

$$= a[(x+1)^2 - x^2] - b(x+1 - x)$$

$$= a[2x+1] - b \text{ — (2)}$$

taking the 2nd difference

$$\Delta^2 y_x = \Delta[a[2x+1] - b]$$

$$= a \Delta[2x+1] - \Delta b$$

$$= 2a \Delta x + a - b$$

$$= 2a[x+1 - x] - a - b$$

$$= 2a + a - b$$

$$= 3a - b$$

$$= a \Delta[2x+1] - \Delta(b) = 0$$

$$= 2a \Delta x + a \Delta 1 = 0$$

$$= 2a \Delta x$$

$$= 2a[x+1 - x]$$

$$= 2a \text{ — (3)}$$

From ②, $\Delta y_x = \frac{1}{2} \Delta^2 y_x - b$

$$\Delta = \frac{1}{2} \Delta^2 y_x - \Delta y_x$$

Then ① $\Rightarrow y = \frac{1}{2} x \Delta^2 y_x - (\frac{1}{2} \Delta^2 y_x - \Delta y_x) x$
 $= \frac{1}{2} (x-1) \Delta^2 y_x$

$$b = \frac{1}{2} (2x+1) \Delta^2 y_x - \Delta y_x$$

$$y_x = \frac{1}{2} \Delta^2 y_x x^2 - [\frac{1}{2} (2x+1) \Delta^2 y_x - \Delta y_x] x$$

$$y_x = \frac{1}{2} \Delta^2 y_x [x^2 - 2x - 1] + \Delta y_x x$$

$$= \frac{1}{2} \Delta^2 y_x (x^2 - 2(x-1)) + \Delta y_x - y_x = y_x$$

or, There are 2 arbitrary constants a and b , taking 1st and 2nd difference;

$$y_x = ax^2 - bx$$

$$y_{x+1} = a(x+1)^2 - b(x+1)$$

$$y_{x+2} = a(x+1)^2 - b(x+2)$$

Elimination arbitrary constants a and b from the given set of eqⁿ.

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix} \begin{vmatrix} y_x & x^2 & x \\ y_{x+1} & (x+1)^2 & (x+1) \\ y_{x+2} & (x+1)^2 & (x+2) \end{vmatrix} = 0$$

$$y_x [(x+1)^2(x+2) - (x+2)^2(x+1)] - y_{x+1} [x^2(x+2) - x(x+2)^2] + y_{x+2} [x^2(x+1) - x(x+1)^2] = 0$$

Find the difference eqⁿ satisfied by the relation.

$$y_n = A \cos n\theta + B \sin n\theta$$

$$y_{n+1} = A \cos (n+1)\theta + B \sin (n+1)\theta$$

$$y_{n+2} = A \cos (n+2)\theta + B \sin (n+2)\theta$$

 $(+)$
 y_n
 $(-)$
 $\cos n\theta$
 $(+)$
 $\sin n\theta$
 $(-)$
 y_{n+1}
 $(+)$
 $\cos (n+1)\theta$
 $(-)$
 $\sin (n+1)\theta$
 $(+)$
 y_{n+2}
 $(-)$
 $\cos (n+2)\theta$
 $(+)$
 $\sin (n+2)\theta$

$$y_n (\cos (n+1)\theta \sin (n+2)\theta - \sin (n+1)\theta \cos (n+2)\theta) - y_{n+1} (\cos n\theta \sin (n+2)\theta - \sin n\theta \cos (n+2)\theta) + y_{n+2} (\cos n\theta \sin (n+1)\theta - \sin n\theta \cos (n+1)\theta) = 0$$

$$y_n [\sin \theta] + y_{n+1} [\sin 2\theta] + y_{n+2} \sin \theta = 0$$

$$y_n \sin \theta - y_{n+1} \sin 2\theta + y_{n+2} \sin \theta = 0$$

$$y_n \sin \theta - y_{n+1} \times 2 \sin \theta \cos \theta + y_{n+2} \sin \theta = 0$$

$$y_n - 2y_{n+1} \cos \theta + y_{n+2} = 0$$

from the difference eqⁿ from the given relation

$$\log \frac{(1+z)}{(1-z)} = y_0 + y_1 z + y_2 z^2 + \dots + y_n z^n + \dots$$

$$\log (1+z) = (1-z) [y_0 + y_1 z + y_2 z^2 + \dots + y_n z^n + \dots]$$

$$z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \frac{z^5}{5} - \dots = (1-z)$$

$$\therefore \log (1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \frac{z^5}{5} - \dots$$

Equating the power on the both sides.

$$z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \frac{z^5}{5} = y_0 + (y_1 - y_0)z + (y_2 - y_1)z^2 + (y_3 - y_2)z^3 + \dots$$

Equating the power of z on both sides.

$$y_1 - y_0 = 1$$

$$y_2 - y_1 = -1/2$$

$$y_3 - y_2 = 1/3$$

$$y_4 - y_3 = 1/4$$

$$\vdots$$

$$y_{n+1} - y_n = \frac{(-1)^n}{(n+1)} \quad \text{or} \quad \Delta y_n = \frac{(-1)^n}{(n+1)}$$

Solution of difference eqⁿ.

A general linear difference eqⁿ of the order r with constant coefficient is given by,

$$[a_0 E^r + a_1 E^{r-1} + a_2 E^{r-2} + \dots + a_{r-1} E + a_r] y_n = F(n)$$

where $a_0, a_1, \dots, a_r$ are constants and $f(n)$ is a function of n and $\Delta = E - 1$, $E y_r = y_{r+1}$.

$$y_{n+r} = E^r y_n$$

Hence any difference eqⁿ may be written in symbolic form as

$$f(E) y_n = F(n)$$

Complete solⁿ of eqⁿ $f(E) y_n = F(n)$ is given by $y = C.F + P.I$

where C.F = complementary function

P.I = particular integral

solve the difference eqⁿ,

$$(\Delta^2 - 3\Delta + 2) y_n = 0$$

$$\Delta = E - 1$$

$$[(E-1)^2 - 3(E-1) + 2] y_n = 0$$

$$(E^2 - 2E + 1 - 3E + 3 + 2) y_n = 0$$

$$(E^2 - 5E + 6) y_n = 0$$

$$(E-3)(E-2) y_n = 0 \implies E = 2, 3.$$

The solⁿ of $Y_n = C.F$

$$Y_n = C_1 2^n + C_2 3^n$$

$$U_{n+3} - 2U_{n+2} - 5U_{n+1} + 6U_n = 0$$

$$Y_{n+r} = E^r Y_n$$

$$U_{n+2} = E^2 U_n$$

$$E^3 U_n - 2E^2 U_n - 5E U_n + 6 = 0$$

$$(E^3 - 2E^2 - 5E + 6) U_n = 0$$

$$(E-1)(E^2 - E - 6) U_n = 0$$

$$(E-1)(E-3)(E+2) U_n = 0$$

$$E = 1, -2, 3.$$

The solⁿ $Y_n = C_1 + C_2 (-2)^n + C_3 3^n$

Solve the difference eqⁿ,

$$Y_{n+4} - 2Y_{n+2} + Y_n = 0$$

$$E^4 Y_n - 2E^2 Y_n + Y_n = 0$$

$$Y_n [E^4 - 2E^2 + 1] = 0$$

$$Y_n [E^2 - 1]^2 = 0$$

$$Y_n (E-1)(E+1)(E-1)(E+1) = 0$$

$$C.F = (C_1 + C_2 n)(-1)^n + (C_3 + C_4 n)(1)^n$$

$$Y_{n+2} - 3Y_{n+1} + 2Y_n = 5^n$$

$$Y_{n+2} - 3Y_{n+1} + 2Y_n = 0$$

$$E^2 Y_n - 3E Y_n + 2Y_n = 0$$

$$(E^2 - 3E + 2) Y_n = 0$$

$$(E-2)(E-1) Y_n = 0$$

$$C.F = C_1 (2)^n + C_2 (1)^n$$

$$= C_2 + C_1 2^n$$

$$P.I = \frac{1}{F(E)} F(n)$$

$$= \frac{1}{(E^2 - 3E + 2)} 5^n$$

$$= \frac{1}{(E-1)(E-2)}$$

$$= \frac{5^n}{(E-1)(E-2)}$$

$$E = 5$$

$$\frac{5^n}{4 \times 3} = \frac{5^n}{12}$$

Solution $C.F + P.I = C_1 + C_2 2^n + \frac{5^n}{12}$

$$U_{n+2} - 7U_{n+1} + 10U_n = 3e^{2n} - 3^n$$

C.F $E^2 U_n - 7 \times E U_n + 10 U_n = 0$

$$U_n [E^2 - 7E + 10] = 0$$

$$U_n [E-5][E-2] = 0$$

$$C_1 [5]^n + C_2 [2]^n = 0$$

P.I $\frac{1}{F(E)} F(n)$

$$= \frac{1}{(E^2 - 7E + 10)} (3e^{2n} - 3^n)$$

$$= \frac{3(e^2)^n}{(E^2 - 7E + 10)} - \frac{3^n}{E^2 - 7E + 10}$$

$$= \frac{3e^{2n}}{(e^4 - 7e + 10)} - \frac{3^n}{(9 - 21 + 10)}$$

$$= \frac{3e^{2n}}{E^4 - 7E + 10} + \frac{3^n}{2} + C_1 [5]^n + C_2 [2]^n$$

$$y_{n+2} + y_{n+1} + 2y_n = \sin 2n$$

$$E^2 y_n + E y_n - 2y_n = \sin 2n$$

$$(E^2 + E - 2) y_n = 0$$

$$(E-1)(E+2) y_n = 0$$

$$\begin{aligned} C.F. &= C_1 (1)^n + C_2 (-2)^n \\ &= C_1 + C_2 (-2)^n \end{aligned}$$

$$P.I. = \frac{1}{(E-1)(E+2)} \sin 2n$$

$$= \left(\frac{e^{2ni} - e^{-2ni}}{2i} \right) \frac{1}{(E^2 + E - 2)}$$

$$= \frac{1}{2i} \left[\frac{(e^{2i})^n - (e^{-2i})^n}{E^2 + E - 2} \right]$$

$$= \frac{1}{2i} \left[\frac{(e^{2i})^n}{(E^2 + E - 2)} - \frac{(e^{-2i})^n}{(E^2 + E - 2)} \right]$$

$$= \frac{1}{2i} \times \frac{(e^{2i})^n}{(e^{4i} + e^{2i} - 2)} - \frac{(e^{-2i})^n}{2i (e^{4i} + e^{-2i} - 2)}$$

$$y_{n+1} - 2y_n = \cos 2n$$

$$(E - 2) y_n = \cos 2n$$

$$E = 2$$

$$C.F. = C_1 (2)^n \text{ --- (1) }$$

$$P.I. = \frac{1}{(E-2)} \cos 2n$$

$$= \frac{1}{(E-2)} \left[\frac{e^{i2n} + e^{-i2n}}{2} \right]$$

$$= \frac{1}{2} \times \left[\frac{(e^{2i})^n + (e^{-2i})^n}{(E-2)} \right]$$

$$= \frac{1}{2} \left[\frac{(e^{2i})^n}{(E-2)} + \frac{(e^{-2i})^n}{e^{-2i} - 2} \right]$$